

Logan Hysen

M.Sc. Candidate – Department of Wildlife
California State Polytechnic University, Humboldt
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Research Interests

My research interests are focused on using spatial methods to understand interactions at the **intersection of wildlife, humans, and the environment**, divided into (1) applied interests, such as understanding the **impact of both global and local disturbances** like climate change and wildfire on species, mapping **movement corridors** to inform management activities, human-wildlife conflict, exploring **habitat selection** for various species, **coupled human and natural systems**, and (2) theoretical interests, such as exploring the inner workings of various modeling approaches to develop best practice recommendations and developing methods to **include more biologically meaningful information in connectivity models**, such as predation risk and human-wildlife interactions.

Education

California State Polytechnic University, Humboldt (formerly Humboldt State University)	May 2023
M.Sc. Wildlife Ecology	
GPA: 4.0/4.0 cumulative	
University of Tennessee, Knoxville, TN	May 2020
B.S. Environmental and Soil Sciences, Environmental Science Concentration	
Watersheds minor	
GPA: 3.96/4.0 cumulative, <i>summa cum laude</i> , Top Upperclass Scholar	

Research Experience

Graduate Student, California State Polytechnic University, Humboldt Department of Wildlife (Arcata, CA)	August 2021 – Present
Advisor: Dr. Ho Yi Wan	
- Performing spatial analyses (e.g., habitat suitability, connectivity modeling) for northern spotted owls using ensemble learning and connectivity software (e.g., Circuitscape, FRAGSTATS) - Investigating the presence-only ensemble habitat selection model development process and developing recommendations for best practices	
Undergraduate Lab Assistant, University of TN, Knoxville Department of Ecology and Evolutionary Biology (Knoxville, TN)	January 2019 – June 2019
- Collected data on larval and adult dung beetles for use in a research study, including monitoring gas exchange and body mass - Assisted research team with tasks necessary to ensure a clean, functioning research environment	
EPA Rain Works competition	Fall 2018
Project Title: <i>Marbled Ecologies</i>	
- Used ArcGIS and Python programming to perform spatial analysis of habitat interconnectivity, stormwater discharge, and pollutant loading for a site on the campus of the University of Tennessee - Coordinated the completion of tasks by members of an interdisciplinary team	

Peer-Reviewed Publications

Google Scholar Link: <https://scholar.google.com/citations?user=QLQRG8kAAAAJ&hl=en>

1. **Hysen, L.**, Nayeri, D., Fogarty, F., Kelly, E., Cushman, S., & Wan, H. Y. (2023) Northern spotted owl nesting habitat under high potential wildfire threats along the California Coastal Redwood Forest. *Science of the Total Environment*. <https://doi.org/10.1016/j.scitotenv.2023.163414>

2. **Hysen, L.**, Nayeri, D., Cushman, S. A., & Wan, H. Y. (2022) Background sampling for multiscale ensemble habitat selection modeling: how many points should be used? *Ecological Informatics*. <https://doi.org/10.1016/j.ecoinf.2022.101914>
3. Nayeri, D., Mohammadi, A., **Hysen, L.**, Hipólito, D., Huber, D., & Wan, H. Y. (2022) Identifying human-caused mortality hotspots to inform human-wildlife conflict mitigation, implications for conservation. *Global Ecology and Conservation*. [10.1016/j.gecco.2022.e02241](https://doi.org/10.1016/j.gecco.2022.e02241)

Submitted and Working Manuscripts

1. Wang, Z., Wang, Y., Cahill Jr., J., Li, X., Wan, H. Y., **Hysen, L.**, Wang, D., & Zhong, Z. In review. Standing litter modifies top-down effects of large herbivores on grassland plant community.
2. **Hysen, L.** & Wan, H. Y. In prep. Impact of barred owl (*Strix varia*) invasion on landscape connectivity for northern spotted owls (*Strix occidentalis caurina*).
3. Ulbing, C., Grubb, B., Taddeo, S., **Hysen, L.**, MacDonald, W., White, E., Fortin, M., Simmons, R., Day, C., & Landguth, E. In prep. Riverscape genetic sampling: what strategy to use and how intense?
4. Nayeri, D., **Hysen, L.**, Willey, D., Ganey, J., Cushman, S., & Wan, H. Y. In prep. Comparing canyonland and forest model for Mexican spotted owl through multi-scale habitat suitability modeling.

Oral Presentations (* denotes invited talks)

1. **Hysen, L.**, Cushman, S., Nayeri, D., & Wan, H. Y. (2023) Sampling the Background: Which strategy should you use? North American Regional Association of the International Association for Landscape Ecology (IALE-NA), Riverside, CA
2. Nayeri, D., **Hysen, L.**, Willey, D., Ganey, J., Cushman, S., & Wan, H. Y. (2023) Characterizing bimodal habitat selection of the Mexican spotted owl with multiscale ensemble modeling. North American Regional Association of the International Association for Landscape Ecology (IALE-NA), Riverside, CA
3. **Hysen, L.**, Nayeri, D., Ganey, J., Cushman, S., & Wan, H. Y. (2022) A tale of two owls: using multi-scale optimization models to identify nuances in habitat selection between two owl subspecies. North American Regional Association of the International Association for Landscape Ecology (IALE-NA), virtual
4. Nayeri, D., **Hysen, L.**, Willey, D., Ganey, J., Cushman, S., & Wan, H. Y. (2022) One species, two worlds: identifying habitat covariates and optimal scales of Mexican spotted owl habitat selection between canyonland and forest habitats. North American Regional Association of the International Association for Landscape Ecology (IALE-NA), virtual
5. ***Hysen, L.** & Nayeri, D. (2022) When two worlds collide: characterizing and predicting the patterns of human-caused mortality of a large carnivore. Conservation Unlimited Tuesday Talks, Cal Poly Humboldt, Arcata, CA
6. **Hysen, L.**, Nayeri, D., Ganey, J., Cushman, S., & Wan, H. Y. (2022) Using multi-scale optimization models to identify differences in habitat selection between two owls. North American Congress for Conservation Biology (NACCB), Reno, NV
7. Nayeri, D., **Hysen, L.**, Willey, D., Ganey, J., Cushman, S., & Wan, H. Y. (2022) Unraveling optimal scales and habitat covariates of Mexican spotted owl habitat selection between two different landscapes. North American Congress for Conservation Biology (NACCB), Reno, NV

Other Professional Positions

Game Pen Technician	September 2021 - Present
Humboldt State University Game Pens (Arcata, CA)	
Supervisor: Jaclyn Patmore	
Field Technician	June 2018 – August 2021
Grow Environmental Solutions, LLC. (Tullahoma, TN)	
Supervisor: Christopher Grow	
• Performed Rare, Threatened, and Endangered species surveys, habitat assessments, mitigation project assessments, wetland delineations, hydrologic determinations, communicated with government agencies • Responsible for compiling reports, technical documents, and associated data	

Funded Grants (Funding to Date: \$1,089.37)

California North Coast Chapter of The Wildlife Society (Co-Principal Investigator)	April 2022
Amount: \$1,089.37	
Research Topic: To conduct a study on demographics of northern spotted owl prey species in partnership with the United States Forest Service	

Unfunded Proposals

Save the Redwoods League Research Grant (<i>Invited to Submit Full Proposal</i>)	September 2021
Amount: \$25,000, unfunded	
Research Topic: To assess fuel loading, habitat, and movement corridors in Northern California	
Yolo Basin Graduate Research Fellowship	June 2021
Amount: \$12,000, unfunded	
Research Topic: To develop wetland connectivity models for avian and amphibian species in the Yolo Basin	
National Science Foundation Graduate Research Fellowship Program	October 2021
Research Topic: To assess forest fuels and develop forest fuel models for northern California to model wildfire risk, unfunded	

Professional Service

Vice President, Wildlife Graduate Student Society, Cal Poly Humboldt	2022-Present
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Awards/Recognitions/Scholarships

University Distinguished Fellowship, Michigan State University (\$106,000.00)	2023-2027
James R. Koplin Outstanding Graduate Student Award, Wildlife Department, Cal Poly Humboldt (\$---	2023
Wildlife Graduate Student Society Mini-Grant, Cal Poly Humboldt (\$400.00)	2023
Student Travel Award – International Association of Landscape Ecology, North America (\$800.00)	2023
Association of Environmental Professionals Bay Area Student Scholarship (\$1,000.00)	2022
Geoffrey C. Bain Jr. Memorial Scholarship, College of Natural Resources & Sciences (\$2,000.00)	2022
Student Travel Award – North American Congress for Conservation Biology (\$250.00)	2022
Student Travel Award – International Association of Landscape Ecology, North America (\$135.00)	2022
Graduate Assistant Tuition Waiver, Wildlife Department, Cal Poly Humboldt (\$7,200.00)	2022
Student Travel Award – Western Section of The Wildlife Society (\$215.00)	2022
Environmental and Soil Sciences Top Upper-class Scholar Award, University of TN, Knoxville (\$200.00)	2019-2020
3rd Place Poster, 7th Annual Watershed Symposium at the University of TN, Knoxville (\$50.00)	2019

Teaching Experience

Graduate Assistant, Ornithology (WLDF 365)	January 2022 – May 2022
Department of Wildlife (Arcata, CA)	
Primary Instructor: Dr. Frank Fogarty III	
Duties included assisting during lab instruction and field trips, helping individual and groups of students understand lab concepts, and grading assignments and exams	
Graduate Assistant, Wildlife Field Techniques (WLDF 311)	August 2022 – December 2022
Department of Wildlife (Arcata, CA)	
Primary Instructor: Dr. Nicholas Kerhoulas	
Duties include assisting during lab instruction and field trips, grading assignments and exams, assist during preparation of lab and field trip material	

Professional Membership

The Wildlife Society	2022 – Present
Northern American Regional Association of the International Association for Landscape Ecology	2022 – Present
Society for Conservation Biology	2022 – Present
California North Coast Chapter of the Wildlife Society	2022 – Present

Skills

Modeling: Multi-scale optimized habitat selection, machine learning (random forest & artificial neural networks), ensemble learning, CDMetaPOP, MaxEnt, statistical models (e.g., GLMM, GLM, GAM), UNICOR, home range estimation, JAGS, PATHWALKER, Circuitscape

GIS: R, Python, Google Earth Engine, FRAGSTATS, ArcGIS

Programming: R, Python, JavaScript, Julia (beginner)

Field: Amphibian and small mammal handling, mist netting, camera trapping, habitat surveys, belt transects, flora & fauna of the U.S. identification, Survey123, wetland delineation

Languages: English (Native), Spanish (Intermediate), Farsi (Intermediate)

Other Experience

Hub Coordinator for Sunrise Movement, Knoxville, TN

January 2019 – July 2020

- Founded an organization focused on climate change outreach to various groups
- Organized events related to environmental protection, communication, inclusivity, and non-violence

References

1. Dr. Jianguo Liu (Ph.D. Major Advisor)
Rachel Carson Chair in Sustainability
Center for Systems Integration and Sustainability
Michigan State University
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2. Dr. Ho Yi Wan (M.Sc. Advisor)
Assistant Professor, Spatial Ecology
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3. Dr. Frank Fogarty III (M.Sc. Committee Member)
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4. Mr. Christopher Grow
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5. Dr. Samuel Cushman
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